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A Novel Framework for Streamlining Well & Surface Integrated Network Model Management in the Oil & Gas Industry

A. Saad, H. Al-Hussain, M Al-Eissa, and A. Al-Abbad, Saudi Aramco, Dhahran, Saudi Arabia

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Abstract

This paper presents a novel framework for an advanced solution to centralize and automate the management and maintenance of well and surface integrated network models (W&SINM). Effective management of intricate W&SINM is crucial to ensure optimal performance, strategic planning, and data-driven decisions in the Oil and Gas industry. It also allows seamless collaboration for multi-disciplinary teams by providing holistic understanding of both surface and sub-surface networks to leverage simulation scenarios, forecasts and predictive analysis.

The solution provides a centralized data repository that integrates with organizational repositories, databases, and modelling software to automate the update of intricate network models. The development methodology capitalizes on analytics of field sensors and surveillance activities, along with advanced modelling technologies to replicate the networks from reservoir to central processing facilities. Integration with existing well model repository, allows automated update of the uploaded network models with latest calibrated well data collected from field sensors. The platform also provides version control and metadata management with customized fine-tuned access control to ensure security.

The development of a centralized network model management repository leads to streamlining the process of modelling update, verification and sharing/collaboration mechanism. The platform also improves integrity and reliability of the models by establishing a transparent and auditable record of all model adjustments and modifications. Additionally, it improves accessibility and collaboration across stakeholders, enhancing communication and facilitating informed decision-making, all while maintaining robust access controls to protect sensitive information. Overall, this platform enables organizations to optimize their modeling processes, reducing errors and increasing productivity, which ultimately leads to better-informed decision-making, improved operational efficiency, and enhanced overall performance.

Introduction

The Oil and Gas industry is a complex and dynamic sector that requires careful planning, optimization, and management to ensure maximum efficiency and productivity. Advanced modelling and simulation techniques are being leveraged to optimize field performance, reduce costs, and inform decision-making. Well & Surface Integrated Network Model (W&SINM) plays a critical role in this context, as it integrates

the different components of the production network, including wells, pipelines, and processing facilities, to provide a holistic understanding of fluids behavior throughout the entire system.

The intricate complexity and large scale of Well and Surface Integrated Network Models (W&SINM), combined with the constantly changing production conditions, necessitate a sophisticated solution to centralize and automate the management and maintenance of these models.

To unlock the full potential of W&SINM, this paper presents a novel framework. The framework offers an advanced solution to centralize and automate W&SINM management and maintenance. This solution includes a centralized data repository that integrates with various organizational systems, such as repositories, databases, and numerical simulation software. The repository automates updates to intricate network models, facilitating effortless collaboration, and reproducibility among stakeholders, while ensuring a secure environment.

Background and System Requirements

The simulation technologies integrating well and surface pipeline network models are mature and well established across the industry (Yu, H. 2024). The research question was how to develop an effective online repository that will facilitate effective model sharing, reproducibility, and automatic update when new data from field sensors emerge, while making sure that it is a secure environment?

The proposed solution must satisfy a set of functional and non-functional requirements to ensure effective management and maintenance of Well and Surface Integrated Network Models (W&SINM). Functionally, the system should enable the upload and download of network models with versioning and integrity checks, allowing users to discover and select wells in bulk within the network model. Additionally, the system should support bulk update jobs, simulation jobs with definable scenario parameters, and online execution against the models, with management of inputs and outputs as artifacts. A job dashboard should provide real-time status updates, reason codes, logs, and access to artifacts, with email notifications sent at job start and finish, including final status and pointers to artifacts. The system should also allow users to save and view scenarios and results. Non-functionally, the system must ensure clear audit trails, tracking all changes made to models, as well as fine-grained access controls for security. Furthermore, the system should be compatible with the chosen toolkit and existing model formats, ensuring seamless integration and minimizing disruptions to existing workflows.

System Design and Architecture

The system is developed using Python as the primary programming language. It consists of several integrated components that provide a user-friendly platform. The platform allows users to easily model, store models, run simulations on the premise cloud, view results, and export data. The main four components of the system are detailed below:

Database

This component serves as a centralized repository, storing a vast array of well test data, simulation output, and other relevant data, providing a single source of truth for all data-related activities. The database is utilized to manage and retrieve data for simulations, leveraging a robust and scalable architecture to handle large volumes of data and ensure high-performance querying and retrieval.

The database is optimized to store a wide range of data types, including numerical, textual, and graphical data, and is designed for efficient data storage and retrieval. Advanced data management features, such as data partitioning, indexing, and caching, are implemented to significantly improve data access and retrieval times.

To ensure the integrity and confidentiality of the stored data, the repository is equipped with robust security features, including data encryption and access control. A data governance framework is also

provided, enabling data quality management, data validation, and data certification, to ensure that the data stored is accurate, complete, and consistent.

Furthermore, the database is integrated with other components of the system, allowing for seamless data exchange and enabling the simulation to leverage the stored data to generate accurate and reliable results. Overall, the data storage component plays a critical role in the overall functionality of the system, providing a reliable and efficient data management infrastructure that supports the simulation and analysis capabilities.

Advance Well Model Repository

This component serves as a repository for calibrated well models, which play a crucial role in updating the network model. The well models are generated automatically from a comprehensive well data documentation, incorporating a wide range of critical details such as well casing and tubing specifications, deviation surveys, fluid composition, and Pressure-Volume-Temperature (PVT) properties of the fluid. Additionally, the well models are informed by well test data, pressure surveys, and well rate test data, which are automatically updated in real-time based on the latest rate test data captured in the field. The accuracy and reliability of these well models are paramount, as they have a direct impact on the network model's performance and decision-making capabilities.

Due to the sensitive nature of the information used to generate these well models, a fine-grain access control system is implemented, ensuring that only authorized users with the necessary clearance can access and utilize the platform. This multi-layered security approach guarantees the confidentiality and integrity of the data, while also enabling authorized users to leverage the platform's capabilities to make informed decisions and optimize well performance. By storing and managing these calibrated well models, this component provides a critical link between data collection, analysis, and decision-making, ultimately driving more efficient and effective well operations.

Advance Surface Network Model Repository

This component serves as a dynamic representation of the pipeline network, providing a comprehensive and up-to-date visualization of the entire system. The pipeline network model is updated in real-time through live synchronization with the well model repository, ensuring that any changes or updates to the well models are immediately reflected in the pipeline network. To accurately model the pipeline network, the component leverages Geographic Information System (GIS) data to retrieve the topology and terrain profiles of the area, allowing for the creation of detailed elevation profiles of the pipelines. This integration of GIS data enables the pipeline network model to account for the complex geography of the area, including hills, valleys, and other terrain features that can impact pipeline flow and pressure. By combining real-time well model data with GIS-based terrain information, the pipeline network model provides a highly accurate and detailed representation of the pipeline system.

User Interface

This component allows users to interact with the system, modify or input data, submit jobs, and visualize results, providing a seamless and intuitive experience for users to engage with the system. The user interface is implemented using a .NET web framework, which provides a flexible and scalable architecture for building robust and user-friendly interfaces. The user interface also provides features such as user authentication, authorization and access control ensuring that the system is secure and that user data is protected.

Figure-1 below demonstrates how the system components interact with each other:

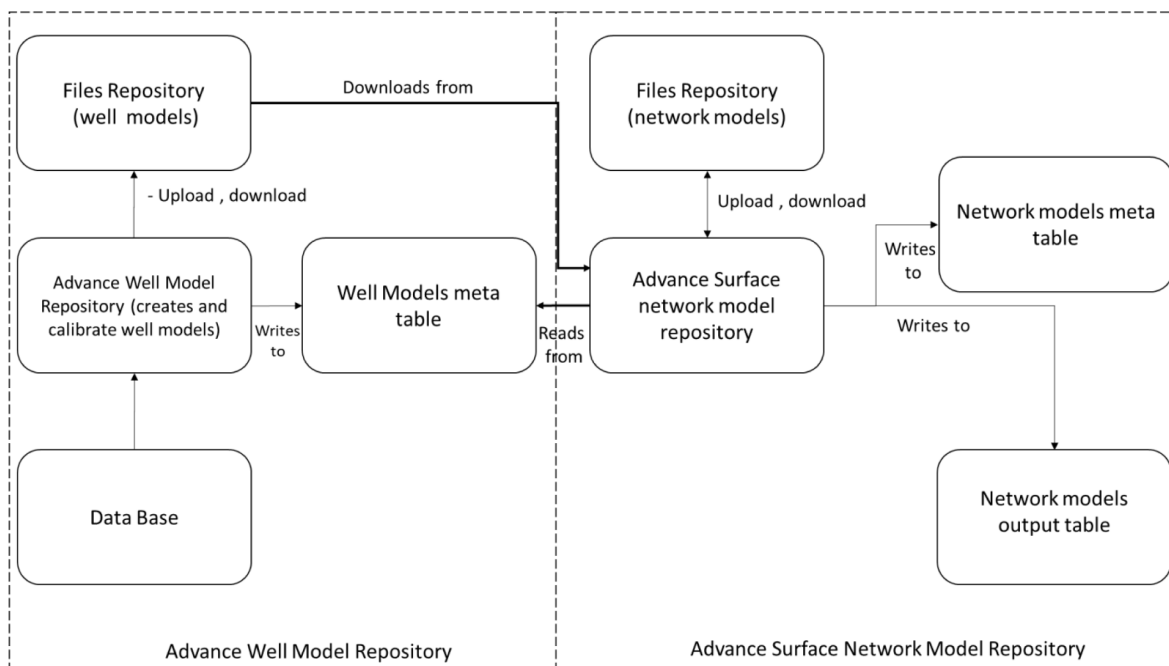


Figure 1—System Architecture

Job-Orchestration Layer

The Job Orchestration Layer is responsible for managing and coordinating the execution of jobs in a scalable and efficient manner. This layer is composed of three primary components: The Control Plane, Execution Plane, and Queuing mechanism. These components work together to receive job requests, prioritize and schedule jobs, execute jobs, and manage job metadata. The following table provides an overview of the key responsibilities of each component, highlighting their roles and interactions within the Job Orchestration Layer:

Table 1—Job-orchestration Layer components and responsibilities

Component	Key responsibilities
Control Plane	- Acts as the entry point for user requests through a .NET Web API. - Authenticates users, maps them to their respective tenant-role, and enforces fine-grained policies. - Validates payload against business rules (quota, time limits, priority). - Writes job-metadata (inputs, priority, timestamps, file references). - Enqueues a lightweight message onto the tenant's queue. - Exposes read-only endpoint for UI: list jobs, filter by status/date.
Execution Plane	- Consists of Python workers running a small Flask service. - Poll the tenant-specific queue, claims the highest-scoring job, and creates an isolated work directory. - Downloads required inputs (which are different for each job, based on user selection). - Stores results or artifacts and link them to job metadata. - Update the Database with job status (Completed/Failed), execution metrics, and artifact pointers. - Emits a JobFinished event on a message broker and trigger notification (email) - Provides health and metrics endpoints for orchestrator auto-scaling
Queuing	- Manages multiple logical sorted set queues per tenant (e.g. Simulation, Update). - Uses a composite score and scheduler to continuously recompute job priorities.

Component	Key responsibilities
	- Workers always process the job with the highest score (i.e. urgent or old jobs first) - Implements visibility-timeout to lock jobs while they're being processed; if a job times out or crash, it re-enters the queue. - After a specified number of re-tries, a job is moved to a dead-letter queue for manual inspection. - Exports queue metrics (length, wait time, throughput) and displays them on operations dashboard

The below chart demonstrates the workflow of how the system update jobs:

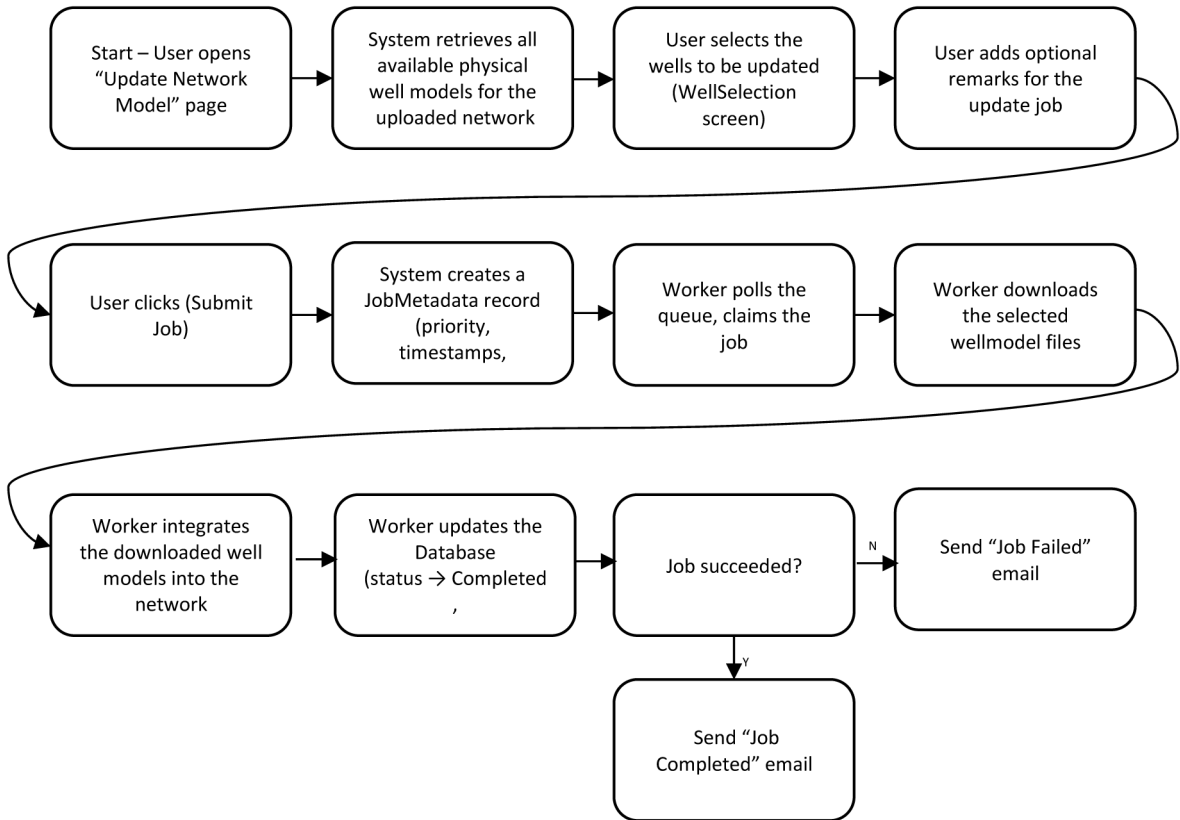


Figure 2—Job-update workflow

System Functionality

The system provides a wide array of tools and functionalities specifically designed to tackle the complexities of W&SINM management and automation. That includes:

Live Synchronization

The platform provides live synchronization, which enables real-time updates and ensures that all data is current and accurate. This functionality allows for the automatic synchronization of data from various sources, including well models, pipeline networks, and other relevant data sources. With live synchronization, users can be confident that they are working with the most up-to-date information, reducing the risk of errors and inconsistencies. This functionality also enables real-time monitoring and alerts, allowing users to respond quickly to changes in the well and pipeline network.

Well Model Bulk Update

The well model update functionality allows for the seamless integration of new well data and ensures that well models are always up-to-date. This functionality enables users to update well models with new data, such as well logs, production data, and other relevant information. The well model update functionality also includes automated data validation and quality control, ensuring that the updated well models are accurate and reliable. With this functionality, users can ensure that their well models are always current and reflective of the latest data, enabling more accurate predictions and optimizations.

New Wells Integration

The new wells integration functionality streamlines the process of adding new wells to the network and ensures that they are properly incorporated into the overall system. This functionality enables users to easily add new wells to the pipeline network, including the creation of new well models and the integration of relevant data. The new wells integration functionality also includes automated workflows and data validation, ensuring that the new wells are properly configured and connected to the existing pipeline network. With this functionality, users can quickly and easily add new wells to the network, reducing the time and effort required to integrate new wells.

Simulation Jobs

The system performs simulation using a numerical simulation software that is integrated into the platform. Allowing the users to run their simulations on the premise cloud without the need to open the simulation software on their local machines. This functionality allows for centralized management of simulation jobs, providing users with access to current and past simulation results. It also allows users to select and input specific parameters for each well in the network model, including gas-oil ratio (GOR), water cut (WC), and liquid rate, while also providing the latest well test data and PVT properties from the field. It allows for the importation of data from external sources, such as Excel. A built-in review feature ensures that users can verify their inputs and settings prior to simulation job submission, thereby minimizing the risk of errors and ensuring the accuracy of simulation results.

Data Retrieval

The system is connected to the database and it retrieves test data and other relevant data automatically without the need for the user to gather and/or export any data to be fed to the simulation models.

Simulation Output Storage

The simulation output storage functionality ensures that all simulation results are properly stored and can be easily retrieved and analyzed. This functionality includes a defined repository in the database for simulation results, enabling users to store and manage large amounts of data. The simulation output storage functionality also includes advanced data management tools, such as data compression and encryption, ensuring that simulation results are secure and reliable. With this functionality, users can ensure that their simulation results are properly stored and can be easily retrieved and analyzed, enabling more accurate predictions and optimizations.

Results Visualization

The system allows the users to visualize the simulation results using a Python-based visualization library. The results are easily accessed through the platform without the need to open the simulation tool and retrieve the results. The results of each run and each model uploaded are saved in the platform, this solves the problem of re-productibility that is often faced by engineers when trying to re-create simulation models.

Background Simulation

The system allows engineers to run the network model through the online platform without opening the modelling software. The modelling software runs in the background, and once the simulation is complete, the engineer is notified via email that the results are ready to be viewed through the system.

Scenario Analysis

The scenario analysis functionality of the system enables users to conduct comprehensive evaluations of various scenarios, facilitating the identification of optimal strategies for enhancing well and pipeline performance. This capability encompasses a range of sensitivity analyses, including pressure, production rate, gas-oil ratio (GOR), water cut (WC), and pipeline size, allowing users to assess the impact of different parameters on system behavior. Furthermore, the system permits users to modify pressure values at distinct points within the pipeline network and analyze the subsequent effects on flow rates, pressures, and other relevant parameters. To facilitate scenario comparison and validation, the system also features a functionality that highlights all user-modified values, enabling users to readily distinguish their scenario-based inputs from well test data and assess the deviations between the two, thereby informing data-driven decision-making and optimization strategies.

Queuing

The queuing system functionality is designed to manage and prioritize simulation jobs, ensuring that multiple users can access the platform simultaneously without compromising performance. This functionality allows users to submit their simulation jobs to a queue, where they are processed in a sequential manner. The queuing system ensures that each simulation job is executed in the order it was received, preventing conflicts and overlaps between different simulations. The system also enables job prioritization, allowing assigning priority levels simulation jobs, ensuring that simulations are executed promptly. Users can also track the status of their simulation jobs in real-time, receiving notifications when their jobs are completed or if any issues arise. The queuing system optimizes resource allocation, ensuring that the platform's computational resources are utilized efficiently and effectively. By implementing a queuing system, the platform provides a robust and scalable solution for managing simulation jobs, enabling multiple users to work efficiently and effectively, while minimizing downtime and maximizing productivity.

Conclusion

In conclusion, the proposed platform presents a comprehensive solution for addressing the complex challenges associated with Well and Integrated Network models (W&INM) Management. By providing a suite of key functionalities, including live synchronization, well model update, new wells integration, simulation execution, data retrieval, simulation output storage, results visualization, background simulation, and scenario analysis, the platform enables users to optimize well and pipeline performance and minimize costs. Furthermore, the platform's advanced simulation and modeling capabilities facilitate the prediction of well and pipeline network behavior under various operating conditions, thereby enabling more accurate predictions and optimizations. The results visualization functionality also enables users to rapidly and intuitively comprehend complex simulation results, thereby reducing the time and effort required for data analysis.

The implementation of this platform will enable its users to make informed, data-driven decisions, enhance operational efficiency, and minimize costs. The platform's scalability, flexibility, and user-friendly interface render it an attractive solution for its purpose. As the industry continues to evolve, the platform's advanced simulation and modeling capabilities will play a critical role in optimizing production processes, reducing environmental footprint, and enhancing overall profitability.

Future Directions

The solution features a robust well and network integrated model that enables scenario-based testing to identify potential back pressure and bottlenecks within the pipelines. This capability allows for the detection and mitigation of limitations that may impact pipeline efficiency. The next phase of development will expand the solution to include advanced pipeline integrity checks, such as corrosion detection and prediction, enabling operators to proactively identify and address potential issues before they become major problems. By leveraging advanced data analytics and machine learning algorithms, the solution will provide real-time insights and recommendations for maintenance and repair.

References

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